

1 Oct 20 HW

Oct 20 homework: Proov 3 equation

$$1. [p_x, \psi(x)] = -i\hbar \frac{\partial \psi}{\partial x}$$

$$2. [p_x^2, \psi(x)] = -2i\hbar \frac{\partial \psi}{\partial x} p_x - \hbar^2 \frac{\partial^2 \psi}{\partial x^2}$$

$$3. [x, y(p(x))] = i\hbar \frac{\partial y(p(x))}{\partial p(x)}$$

proof now:

$$1. [p_x, \psi] \psi = p_x [\psi^2] - \psi p_x [\psi] = -i\hbar 2\psi \partial_x \psi + i\hbar \psi \partial_x \psi = -i\hbar \partial_x \psi \psi$$

$$2. [p_x^2, \psi] = -\hbar^2 \partial_x \partial_x \psi^2 + \hbar^2 \psi \partial_x \partial_x \psi = -2\hbar i \partial_x \psi \times p_x \psi - \hbar^2 \psi \partial_x \partial_x \psi =$$

$$(-2\hbar i \partial_x \psi p_x - \hbar^2 \partial_x \partial_x \psi) \psi$$

$$3. [x, y(p(x))] \text{ and } y(p(x)) = \sum_n C_n p(x)^n \text{ so } \sum_n C_n [x, p_x^n] = [x, y(p(x))] \text{ and}$$

$$[x, p_x^n] \psi = x \times p_x^n \psi - p_x^n (x \psi) = i\hbar n p_x^{n-1} \psi = \sum_n C_n [x, p_x^n] = [x, \sum_n C_n p_x^n] = i\hbar \sum_n C_n n p_x^{n-1} \psi =$$

$$i\hbar \partial_{p_x} y(p_x) \psi$$

$$p_x^n (x \psi) = p_x^{n-1} (c \psi + x \partial_x \psi) = c p_x^{n-1} \psi + p_x^{n-2} (c \partial \psi + x p_x^2 \psi) \text{ so } = n c p_x^{n-1} \psi +$$

$$p_x^n \psi \text{ and } c = -i\hbar$$

$$p_x [x \psi] = -i\hbar \psi + x p_x \psi$$

D O N E .

2 Oct 23HW part2

PROOF: $e^{a \frac{d}{dt}} \psi = \psi(x+a)$

$$e^{a \frac{d}{dt}} \psi = \sum_n \frac{a^n}{n!} \frac{d^n}{dt^n} \psi; \psi(x+y) = \sum_n \frac{\frac{d^n \psi}{dt^n} |_{y=0}}{n!} y^n \text{ consider } y=a: \sum_n \frac{\frac{d^n \psi(x)}{dt^n}}{n!} a^n =$$

$$\sum_n \frac{a^n}{n!} \frac{d^n}{dt^n} = e^{a \frac{d}{dt}} \psi$$

DONE.

3 Oct 28HW

A:

$$\text{let } l_{\pm} = l_x \pm i l_y$$

PROOF:

$$(a). [l_z, l_{\pm}] = \pm \hbar l_{\pm}$$

$$(b). l_{\pm} l_{\mp} = l^2 - l_z^2 \pm \hbar l_z$$

$$(c). [l_+, l_-] = 2\hbar l_z$$

proof a:

$$[l_z, l_{\pm}] = [l_z, l_x] \pm i [l_z, l_y] = i\hbar l_y \pm i(-i\hbar l_x) = i\hbar l_y \pm \hbar l_x = \pm \hbar l_{\pm}$$

proof b:

$$l_{\pm} l_{\mp} = (l_x \pm i l_y)(l_x \mp i l_y) = l_x^2 \mp i l_x l_y \pm i l_y l_x + l_y^2 = l_x^2 + l_y^2 \pm i(l_y l_x - l_x l_y) =$$

$$l^2 - l_z^2 \pm i [l_y, l_x] = l^2 - l_z^2 \pm \hbar l_z$$

proof c:

$$l_+ l_- - l_- l_+ = (l_x + i l_y)(l_x - i l_y) - (l_x - i l_y)(l_x + i l_y) = -i l_x l_y + i l_y l_x - i l_x l_y +$$

$$i l_y l_x = 2i [l_y, l_x] = 2\hbar l_z$$

B:

PROOF:

$$(a) \ p \times l + l \times p = 2i\hbar p$$

$$(b) \ i\hbar(p \times l - l \times p) = [l^2, p]$$

proof a:

$$\begin{aligned} & (p_1 l_2 - p_2 l_1)k + (p_2 l_3 - p_3 l_2)i + (p_3 l_1 - p_1 l_3)j + (l_1 p_2 - l_2 p_1)k + (l_2 p_3 - l_3 p_2)i + \\ & (l_3 p_1 - l_1 p_3)j = \\ & ([p_1, l_2] + [l_1, p_2])k + ([p_2, l_3] + [l_2, p_3])i + ([p_3, l_1] + [l_3, p_1])j \\ & (i2\hbar p_3)k + (i2\hbar p_1)i + (i2\hbar p_2)j = i2\hbar \mathbf{p} \end{aligned}$$

proof b:

$$\begin{aligned} i\hbar(p \times l - l \times p) &= (p_1 l_2 - p_2 l_1 - l_1 p_2 + l_2 p_1)k + (p_2 l_3 - p_3 l_2 - l_2 p_3 + l_3 p_2)i + \\ & (p_3 l_1 - p_1 l_3 - l_3 p_1 + l_1 p_3)j \\ [l^2, p] &= [l_x^2 + l_y^2 + l_z^2, p] \\ [l_x^2, p] &= l_x[l_x, p] + [l_x, p]l_x = l_x(i\hbar(p_z j - p_y k)) + (i\hbar(p_z j - p_y k))l_x = i\hbar(l_x p_z + \\ & p_z l_x)j - i\hbar(l_x p_y + p_y l_x)k \\ [l_y^2, p] &= l_y[l_y, p] + [l_y, p]l_y = l_y(i\hbar(p_x k - p_z i)) + (i\hbar(p_x k - p_z i))l_y = i\hbar(l_y p_x + \\ & p_x l_y)k - i\hbar(l_y p_z + p_z l_y)i \\ [l_z^2, p] &= l_z[l_z, p] + [l_z, p]l_z = l_z(i\hbar(p_y i - p_x j)) + (i\hbar(p_y i - p_x j))l_z = i\hbar(l_z p_y + \\ & p_y l_z)i - i\hbar(l_z p_x + p_x l_z)j \\ [l^2, p] &= i\hbar(l_y p_x + p_x l_y - l_x p_y - p_y l_x)k + i\hbar(l_z p_y + p_y l_z - l_y p_z - p_z l_y)i + \\ & i\hbar(l_x p_z + p_z l_x - l_z p_x - p_x l_z)j \\ & \text{so we have } i\hbar(p \times l - l \times p) = [l^2, p] \end{aligned}$$

4 Oct 30 HW

$$\psi_n = A_n e^{-\alpha^2 x^2 / 2} H_n(\alpha x)$$

2.7

(1)

$$H_n(\alpha x) = \frac{1}{2\alpha x}(H_{n+1}(\alpha x) + 2nH_{n-1}(\alpha x))$$

$$x\psi_n = \frac{1}{2\alpha}(H_{n+1}(\alpha x) + 2nH_{n-1}(\alpha x))A_n e^{-\alpha^2 x^2 / 2}$$

$$A_n = \frac{1}{\sqrt{2n}} A_{n-1}; A_n = \sqrt{2n+2} A_{n+1}$$

$$\text{so we get: } x\psi_n = \frac{1}{2\alpha}(A_n H_{n+1}(\alpha x) + 2nA_n H_{n-1}(\alpha x))e^{-\alpha^2 x^2 / 2}$$

$$\frac{1}{2\alpha}(\sqrt{2(n+1)}A_{n+1}(\alpha x) + 2n\frac{1}{\sqrt{2n}}A_{n-1}H_{n-1}(\alpha x))e^{-\alpha^2 x^2 / 2}$$

$$= \frac{1}{2\alpha}(\sqrt{2(n+1)}\psi_{n+1} + \sqrt{2n}\psi_{n-1})$$

(2)

$$x^2\psi_n = \frac{1}{2\alpha}(\sqrt{2(n+1)}x\psi_{n+1} + \sqrt{2n}x\psi_{n-1})$$

$$\begin{aligned} x^2\psi_n &= \frac{1}{2\alpha}(\sqrt{2(n+1)}[\frac{1}{2\alpha}(\sqrt{2(n+2)}\psi_{n+2} + \sqrt{2(n+1)}\psi_n)] + \sqrt{2n}[\frac{1}{2\alpha}(\sqrt{2n}\psi_n + \\ & \sqrt{2(n-1)}\psi_{n-2})]) = \frac{1}{4\alpha^2}[\sqrt{(4(n+1)(n+2))}\psi_{n+2} + 2(n+1)\psi_n + 2n\psi_n + \sqrt{4n(n-1)}\psi_{n-2}] \\ &= \frac{1}{2\alpha^2}[\sqrt{(n+1)(n+2)}\psi_{n+2} + (2n+1)\psi_n + \sqrt{n(n-1)}\psi_{n-2}] \end{aligned}$$

(3)

$$\bar{x} = \int \psi_n^* x \psi_n = 0$$

$$V = \frac{1}{2}Kx^2; \bar{V} = \int \psi_n^* V \psi_n = \frac{1}{2}K \int \psi_n^* x^2 \psi_n = \frac{1}{2}K \frac{1}{2\alpha^2}(2n+1) = \frac{K(2n+1)}{4\alpha^2} =$$

$$\frac{\omega(2n+1)\hbar}{4} = \frac{E_n}{2}$$

2.8

$$H'_n(\alpha z) = 2n\alpha H_{n-1}(\alpha z)$$

$$\begin{aligned}
\psi'_n &= A_n e^{-\alpha^2 x^2/2} 2n\alpha H_{n-1}(\alpha x) - \alpha^2 x A_n e^{-\alpha^2 x^2/2} \left[\frac{1}{2\alpha x} (H_{n+1} + 2nH_{n-1}) \right] \\
&= 2n\alpha A_n e^{-\alpha^2 x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2 x^2/2} H_{n+1} - \alpha n A_n e^{-\alpha^2 x^2/2} H_{n-1} \\
&= \alpha n A_n e^{-\alpha^2 x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2 x^2/2} H_{n+1} \\
A_n &= \frac{1}{\sqrt{2n}} A_{n-1}; A_n = \sqrt{2n+2} A_{n+1} \\
&= \alpha \sqrt{\frac{n}{2}} \psi_{n-1} - \alpha \sqrt{\frac{n+1}{2}} \psi_{n+1}
\end{aligned}$$

(2)

$$\begin{aligned}
\frac{d}{dx} \psi'_n &= \alpha \left[\sqrt{\frac{n}{2}} \psi'_{n-1} - \sqrt{\frac{n+1}{2}} \psi'_{n+1} \right] \\
\frac{d}{dx} \psi'_n &= \alpha \left[\sqrt{\frac{n}{2}} \left[\alpha \sqrt{\frac{n-1}{2}} \psi_{n-2} - \alpha \sqrt{\frac{n}{2}} \psi_n \right] - \sqrt{\frac{n+1}{2}} \left[\alpha \sqrt{\frac{n+1}{2}} \psi_n - \alpha \sqrt{\frac{n+2}{2}} \psi_{n+2} \right] \right] \\
\frac{d}{dx} \psi'_n &= \alpha^2/2 \left[\sqrt{(n-1)n} \psi_{n-2} - n \psi_n - (n+1) \psi_n + \sqrt{(n+1)(n+2)} \psi_{n+2} \right] \\
\frac{d}{dx} \psi'_n &= \alpha^2/2 \left[\sqrt{(n-1)n} \psi_{n-2} - (2n+1) \psi_n + \sqrt{(n+1)(n+2)} \psi_{n+2} \right]
\end{aligned}$$

(3)

$$\begin{aligned}
\bar{p} &= \int \psi_n^* p \psi_n = 0 \\
\bar{T} &= \int \psi_n^* T \psi_n = \int \psi_n^* (-\hbar^2/(2m)) \frac{d^2}{dx^2} \psi_n = \frac{\hbar^2}{2m} \int \psi_n^* \frac{d^2}{dx^2} \psi_n = \frac{\hbar^2}{2m} \left[-\frac{\alpha^2}{2} (2n+1) \right] \\
&= \frac{\hbar \alpha^2}{4m} (2n+1) = \frac{\hbar \omega}{4} (2n+1) = \frac{1}{2} E_n
\end{aligned}$$

2.9

$$\Delta x \Delta p = \sqrt{x^2 p^2} = \sqrt{E_n m E_n / K} = \frac{E_n}{\omega} = \frac{\hbar}{2} (2n+1)$$

3.14

$$\begin{aligned}
l_z \text{ Eigenstate: } \psi_m &= \sqrt{\frac{1}{2\pi}} e^{im\phi} \\
\int \phi^* l_z \phi &= \int (l_z \phi)^* \phi \\
\int \phi^* [l_y, l_z] \phi &= \int \phi^* l_y l_z \phi - \phi^* l_z l_y \phi = \int \phi^* l_y \phi \bar{h} m - (l_z \phi)^* l_y \phi = \bar{h} m \bar{l}_y - \bar{h} m \bar{l}_y = 0 \\
&= i \hbar l_x \\
\int \phi^* [l_z, l_x] \phi &= \int \phi^* l_z l_x \phi - \phi^* l_x l_z \phi = \int (l_z \phi)^* l_x \phi - (l_x \phi)^* l_z \phi = \int \phi^* \bar{h} m l_x \phi - \bar{h} m \phi^* l_x \phi = 0 = i \hbar l_y
\end{aligned}$$

5 Nov 4 HW

3.15

$$\begin{aligned}
\psi &= Y_{lm} \text{ so } \bar{l}_x^2 = \int \psi^* l_x^2 \psi d\Omega \text{ and } l_x^2 = \frac{1}{i\hbar} [l_y, l_z] l_x = \frac{1}{i\hbar} (l_y l_z - l_z l_y) l_x = \\
&\frac{1}{i\hbar} (l_y l_z l_x - l_z l_y l_x) \text{ and we know } l_z l_x = l_x l_z + i \hbar l_y \text{ so } : \frac{1}{i\hbar} (l_y l_x l_z + i \hbar l_y^2 - l_z l_y l_x) = \\
&\frac{1}{i\hbar} l_y l_x l_z + l_y^2 - \frac{1}{i\hbar} l_z l_y l_x = l_x^2 \\
\text{so } \bar{l}_x^2 &= \int \psi^* \left(\frac{1}{i\hbar} l_y l_x l_z + l_y^2 - \frac{1}{i\hbar} l_z l_y l_x \right) \psi d\Omega = \bar{l}_y^2 + \int \psi^* \frac{1}{i\hbar} l_x l_z \bar{h} m \psi - \frac{1}{i\hbar} \bar{h} m \psi^* l_y l_x \psi d\Omega = \bar{l}_y^2 \\
\text{so we have } \bar{l}_x^2 &= \bar{l}_y^2 \text{ now} \\
\text{we know } l^2 &= l_x^2 + l_y^2 + l_z^2, \text{ so } \bar{l}^2 = 2\bar{l}_x^2 + \bar{l}_z^2 \\
\bar{l}_x^2 &= \frac{1}{2} (\int \psi^* l^2 \psi - \psi^* l_z^2 \psi d\Omega) = \frac{1}{2} (l(l+1)\hbar^2 - m^2\hbar^2) \\
\text{so } (\Delta \bar{l}_x)^2 &\text{ we have } : \Delta l_x = \sqrt{-\bar{l}_x^2 + \bar{l}_x^2} = \sqrt{\bar{l}_x^2} \text{ and thus } (\Delta l_x)^2 = |\bar{l}_x^2| = \\
&\frac{1}{2} (l(l+1)\hbar^2 - m^2\hbar^2) \text{ so we have} \\
(\Delta \bar{l}_x)^2 &= \frac{1}{2} (l(l+1)\hbar^2 - m^2\hbar^2) = (\Delta \bar{l}_y)^2
\end{aligned}$$

3.16

$$\text{set } \psi = c_1 Y_{11} + c_2 Y_{20}$$

$l_z \psi = c_1 \bar{h} Y_{11} + c_2 \times 0 = c_1 \bar{h} Y_{11}$ so the possibility of status $0\bar{h}$ is $|c_2|^2$, and $\bar{h}$ is 1, so $l_z = \bar{h}$ and the average $= |c_1|^2 \bar{h}$

$l^2 \psi = c_1 \times 1 \times 2 Y_{11} \bar{h}^2 + c_2 \times 2 \times 3 Y_{20} \bar{h}^2$ so l^2 can be $2\bar{h}^2$ possibility is $|c_1|^2$, and $6\bar{h}^2$ possibility is $|c_2|^2$.

6 Nov 6 HW

*3.18

$$\begin{aligned} e^{-aA} B e^{aA} &= \Sigma \frac{-a^n A^n}{n!} B \Sigma \frac{a^m A^m}{m!} = \Sigma \Sigma \frac{-a^n A^n}{n!} B \frac{a^m A^m}{m!} = \Sigma_{m,n} \frac{-1^n a^{m+n} A^n B A^m}{n! m!} \\ &= \Sigma_{m,n} \frac{-1^n a^{m+n} A^n B A^m}{n! m!} = B - aAB + aBA - a^2ABA + a^2BA^2/2 + a^2A^2B/2... \\ &= B - a[A, B] - \frac{1}{2}a^2(A[A, B])... \text{ for any } m+n=i \text{ we can have} \\ &a^i \Sigma_{m+n=i} \frac{-1^n A^n B A^m}{n! m!} \text{ for } i=3 \text{ we can have: } a^3 \Sigma_{m+n=3} \frac{-1^n A^n B A^m}{n! m!} \\ &= a^3 [[A, [A, B]], A]/6 = -\frac{a^3}{3!} [A, [A, [A, B]]] \end{aligned}$$

$$\text{Now set for } i = 0, 1, 2, 3 \text{ we set } F_i = \frac{-a}{i+1} [A, F_{i-1}] = \Sigma_{m+n=i} \frac{a^i (-1)^n A^n B A^m}{m! n!}$$

Now consider for ant k satisfied F_k satisfied : $F_k = \frac{a^k}{k!} (-1)^k [A, [A, [A, \dots [A, B]]]] = \Sigma_{m+n=k} \frac{a^k (-1)^n A^n B A^m}{m! n!}$ and for k+1 we have:

$$\begin{aligned} -a/(k+1) [A, F_k] &= \frac{a^{k+1}}{(k+1)!} (-1)^{k+1} [A, [A, \dots [A, B]]] = \Sigma_{m+n=k} \frac{a^{k+1} (-1)^n [A, A^n B A^m]}{m! n! (k+1)} \\ \text{and for } [A, A^n B A^m] &= [A, A^n] B A^m + A^n [A, B A^m] = [A, A^n] B A^m + A^n [A, B] A^m + \\ A^n B [A, A^m] &= A^n [A, B] A^m = A^{n+1} B A^m - A^n B A^{m+1} = A^{n+1} B A^m + (-1) A^n B A^{m+1} \text{ so} \\ \Sigma_{m+n=k} \frac{a^{k+1} (-1)^n (A^{n+1} B A^m + (-1) A^n B A^{m+1})}{m! n! (k+1)} \\ \Sigma_{m+n=k} \frac{a^{k+1} (-1)^{n+1} A^{n+1} B A^m + a^{k+1} (-1)^n A^n B A^{m+1}}{m! n! (k+1)} &= \Sigma_{m+n'=k+1} \frac{a^{k+1} (-1)^{n'} A^{n'} B A^m}{m! (n'-1)! (k+1)} + \\ \Sigma_{m'+n=k+1} \frac{a^{k+1} (-1)^n A^n B A^{m'}}{(m'-1)! n! (k+1)} - \frac{a^{k+1} A B A^k}{(k+1)!} - \frac{a^{k+1} (-1)^k A^k B A}{(k+1)!} &= \Sigma_{m+n'=k+1} n' \frac{a^{k+1} (-1)^{n'} A^{n'} B A^m}{m! (n')! (k+1)} + \\ \Sigma_{m'+n=k+1} m' \frac{a^{k+1} (-1)^n A^n B A^{m'}}{(m')! n! (k+1)} + \frac{a^{k+1} A B A^k}{(k+1)!} - \frac{a^{k+1} (-1)^k A^k B A}{(k+1)!} & \text{ 1 and 3 combine can} \\ \text{have a } m \neq 1 \text{ term and 2 and 4 combine can have a } n \neq 1 \text{ term : } \Sigma_{m+n'=k+1; m \neq 1} n' \frac{a^{k+1} (-1)^{n'} A^{n'} B A^m}{m! (n')! (k+1)} + \\ \Sigma_{m'+n=k+1; n \neq 1} m' \frac{a^{k+1} (-1)^n A^n B A^{m'}}{(m')! n! (k+1)} & \text{ so we get satisfied for full } k+1 \text{ terms : } m=0, 1, 2, \dots, k \\ \text{total } k+1, \text{ and we set } m+n=k+1, \text{ we get: } \Sigma \frac{a^{k+1} (-1)^n A^n B A^m}{m! n!} \end{aligned}$$

PROOF DONE.

7 Nov 13

4.4: proof: $-\hbar^2 \frac{d^2}{dt^2} \bar{A} = \overline{[A, H], H}$

PROOF: $\frac{d}{dt} \bar{C} = \frac{1}{i\hbar} [\bar{C}, \bar{B}]$ and let $\bar{C} = \frac{d}{dt} \bar{A} = \frac{1}{i\hbar} [\bar{A}, \bar{H}]$ so

$$\frac{d}{dt} \frac{d}{dt} \bar{A} = -\frac{1}{\hbar^2} \overline{[A, H], H}$$

PROVED

4.5: proof: $\frac{d}{dt} \bar{A} = 0$ $\bar{A}$ is under a stationary state.

$$\text{PROVE: } \frac{d}{dt} \bar{A} = \frac{1}{i\hbar} [\bar{A}, \bar{H}] = \frac{1}{i\hbar} \int \psi^* A H \psi - \psi^* H A \psi = \frac{1}{i\hbar} (E \int \psi^* A \psi - E \int \psi^* A \psi) =$$

0

DONE

8 Nov 18

homework:

4.2:

(a)

$$H = h(q_1) + h(q_2)$$
$$q_1, q_2 = \phi_1, \phi_1; \phi_1, \phi_2; \phi_1, \phi_3; \phi_2, \phi_2; \phi_2, \phi_3; \phi_3, \phi_3.$$

6 kinds

(b)

$$H = h(q_1) + h(q_2)$$
$$q_1, q_2 = \phi_1, \phi_2, \phi_2, \phi_1; \phi_1, \phi_3; \phi_3, \phi_1; \phi_2, \phi_3; \phi_3, \phi_2.$$

3 kinds

(c)

$$3 \times 3 = 9$$

9 kinds

4.3:

(a)

$$3^3 = 27$$

27 kinds

(b)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3} :$$
$$1, 2, 3$$

1 kind

(c)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3} :$$
$$1, 1, 1; 1, 1, 2; 1, 1, 3; 1, 2, 2; 1, 2, 3;$$
$$1, 3, 3; 2, 2, 2; 2, 2, 3; 2, 3, 3; 3, 3, 3;$$

10 kinds

4.8:

proof:

$$(\frac{dF}{dt})_{nn'} = i\omega_{nn'}F_{nn'}, \omega_{nn'} = (E_n - E_{n'})/\hbar \tag{1}$$

prove:

$$\begin{aligned}
\frac{dF}{dt} &= \frac{1}{i\hbar}[F(t), H] \\
\left(\frac{dF}{dt}\right)_{nn'} &= \frac{1}{i\hbar} \int \phi_n F H \phi_{n'} - \phi_n H F \phi_{n'} \\
&= \frac{1}{i\hbar} (E_{n'} \int \phi_n F \phi_{n'} - E_n \int \phi_n F \phi_{n'}) \\
&= \frac{1}{i\hbar} (E_{n'} - E_n) \int \phi_n F \phi_{n'} \\
&= -\frac{1}{i\hbar} \hbar \omega_{nn'} F_{nn'} \\
&= i\omega_{nn'} F_{nn'}
\end{aligned}$$

9 Nov 20

4.7: proof:

$$\frac{\partial E_n}{\partial \lambda} = \langle \psi_n | \frac{\partial H}{\partial \lambda} | \psi_n \rangle \quad (2)$$

PROVE:

$$\begin{aligned} E_n \psi_n &= \hat{H} \psi_n \\ \psi_{na}^* E_n \psi_n^a &= E_n = \psi_{na}^* \hat{H} \psi_n^a \\ \frac{\partial E_n}{\partial \lambda} &= \frac{\partial}{\partial \lambda} (\psi_{na}^* \hat{H} \psi_n^a) \\ &= \psi_{na}^* \frac{\partial \hat{H}}{\partial \lambda} \psi_n^a = \langle \psi_n | \frac{\partial \hat{H}}{\partial \lambda} | \psi_n \rangle \end{aligned}$$

PROVED

10 Nov 25

$\psi = c_1 Y_{11} + c_2 Y_{20}$ try to solve all result and probablility of $\hat{l}_x \psi$

SOLVE:

$$\psi = 0 \times Y_{1-1} + 0 \times Y_{10} + c_1 Y_{11} + 0 \times Y_{2-2} + 0 \times Y_{2-1} + c_1 Y_{20} + 0 \times Y_{21} + 0 \times Y_{22}$$

$$\psi^a = \psi^i \phi_i^a$$

$$l_{xb}^a = \frac{1}{2} l_{-b}^a + \frac{1}{2} l_{+b}^a$$

$$l_{xj}^i = \phi_a^{*i} l_{xb}^a \phi_j^b$$

for $l=1$

$$Y_{1-1} l_x Y_{1-1} = Y_{1-1} \frac{1}{2} (l_+ + l_-) Y_{1-1} = \frac{1}{2} (Y_{1-1} a_{1-1} Y_{10} + Y_{1-1} b_{1-1} \times 0) = 0$$

$$Y_{1-1} l_x Y_{10} = \frac{1}{2} (Y_{1-1} a_{10} Y_{11} + Y_{1-1} b_{10} Y_{1-1}) = \frac{1}{2} \sqrt{l(l+1) - m(m-1)} = \frac{1}{2} \sqrt{2}$$

$$Y_{1-1} l_x Y_{11} = 0$$

$$Y_{10} l_x Y_{1-1} = \frac{1}{2} (Y_{10} a_{1-1} Y_{10}) = \frac{1}{2} \sqrt{l(l+1) - m(m+1)} = \frac{1}{2} \sqrt{2}$$

$$Y_{10} l_x Y_{10} = 0$$

$$Y_{10}l_xY_{11} = \frac{1}{2}(Y_{10}a_{11} \times 0 + Y_{10}b_{11}Y_{10}) = \frac{1}{2}\sqrt{2}$$

$$Y_{11}l_xY_{1-1} = 0$$

$$Y_{11}l_xY_{10} = \frac{1}{2}(Y_{11}a_{10}Y_{11}) = \frac{1}{2}\sqrt{2}$$

for any $l=2$

$$Y_{2-2}l_xY_{2-2} = 0$$

$$Y_{2-2}l_xY_{2-1} = \frac{1}{2}Y_{2-2}b_{2-1}Y_{2-2} = \frac{1}{2}\sqrt{6 - (-1)(-2)} = 1$$

$$Y_{2-2}l_xY_{20} = 0$$

$$Y_{2-1}l_xY_{2-2} = \frac{1}{2}(Y_{2-1}a_{2-2}Y_{2-1}) = \frac{1}{2}\sqrt{6 - (-2)(-1)} = 1$$

$$Y_{2-1}l_xY_{20} = \frac{1}{2}(Y_{2-1}b_{20}Y_{2-1}) = \frac{1}{2}\sqrt{6} = \frac{\sqrt{6}}{2}$$

$$Y_{20}l_xY_{2-1} = \frac{1}{2}(Y_{20}a_{2-1}Y_{20}) = \frac{1}{2}\sqrt{6} = \frac{\sqrt{6}}{2}$$

$$Y_{20}l_xY_{21} = \frac{1}{2}Y_{20}b_{21}Y_{20} = \frac{1}{2}\sqrt{6} = \frac{\sqrt{6}}{2}$$

$$Y_{21}l_xY_{20} = \frac{1}{2}(Y_{21}a_{20}Y_{21}) = \frac{1}{2}\sqrt{6}$$

$$Y_{21}l_xY_{22} = \frac{1}{2}(Y_{21}b_{22}Y_{21}) = \frac{1}{2}\sqrt{6 - 2} = 1$$

$$Y_{22}l_xY_{21} = \frac{1}{2}(Y_{22}a_{21}Y_{22}) = \frac{1}{2}\sqrt{6 - 2} = 1$$

we have :

$$l_{x2}^1 = l_{x1}^2 = l_{x3}^2 = l_{x2}^3 = \frac{1}{\sqrt{2}}\hbar$$

$$l_{x5}^4 = l_{x4}^5 = l_{x8}^7 = l_{x7}^8 = \hbar$$

$$l_{x6}^5 = l_{x5}^6 = l_{x7}^6 = l_{x6}^7 = \frac{\sqrt{6}}{2}\hbar$$

$$l_{xj}^i = \begin{bmatrix} 0 & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & 0 & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & -\lambda & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & -\lambda \end{vmatrix} = 0$$

$$\lambda(\lambda^2 - \frac{\hbar^2}{2}) - \frac{\hbar^2}{2}\lambda = 0$$

$$\lambda_1 = 0; \lambda_2 = \hbar; \lambda_3 = -\hbar$$

so eigenstate : ξ

$$\begin{bmatrix} 0 & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & 0 & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & 0 \end{bmatrix} \times \vec{\xi}_0 = 0; \xi_0^2 = 0; \xi_0^1 = -\xi_0^3;$$

$$\vec{\xi}_0 = \begin{bmatrix} k \\ 0 \\ -k \end{bmatrix}$$

$$\begin{bmatrix} \hbar & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & \hbar & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & \hbar \end{bmatrix} \times \vec{\xi}_{-1} = 0; \xi_{-1}^1 + \frac{1}{\sqrt{2}}\xi_{-1}^2 = 0; \frac{1}{\sqrt{2}}\xi_{-1}^2 + \xi_{-1}^3 = 0; \xi_{-1}^1 = \xi_{-1}^3 = -\frac{1}{\sqrt{2}}\xi_{-1}^2$$

$$\vec{\xi}_{-1} = \begin{bmatrix} k \\ -\sqrt{2}k \\ k \end{bmatrix}$$

$$\begin{bmatrix} -\hbar & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & -\hbar & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & -\hbar \end{bmatrix} \times \vec{\xi}_1 = 0; \xi_1^1 = \frac{1}{\sqrt{2}}\xi_1^2; \frac{1}{\sqrt{2}}\xi_1^2 = \xi_1^3; \xi_1^1 = \xi_1^3 = \frac{1}{\sqrt{2}}\xi_1^2$$

$$\vec{\xi}_1 = \begin{bmatrix} k \\ \sqrt{2}k \\ k \end{bmatrix}$$

the posibility:

$$P_0 = (\phi_a^{*3}\xi_0^a)^2 = \frac{1}{2}$$

$$P_{-1} = (\phi_a^{*3}\xi_{-1}^a)^2 = \frac{1}{4}$$

$$P_1 = (\phi_a^{*3}\xi_1^a)^2 = \frac{1}{4}$$

now consider the $l=2$

$$\begin{aligned}
& \begin{bmatrix} 0 & \hbar & 0 & 0 & 0 \\ \hbar & 0 & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \hbar \\ 0 & 0 & 0 & \hbar & 0 \end{bmatrix} \\
& \begin{vmatrix} -\lambda & \hbar & 0 & 0 & 0 \\ \hbar & -\lambda & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -\lambda & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & -\lambda & \hbar \\ 0 & 0 & 0 & \hbar & -\lambda \end{vmatrix} = 0 \\
& \begin{vmatrix} -\lambda & 0 & 0 & 0 & 0 \\ \hbar & \frac{\hbar^2}{\lambda} - \lambda & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -\lambda & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & \frac{\hbar^2}{\lambda} - \lambda & \hbar \\ 0 & 0 & 0 & 0 & -\lambda \end{vmatrix} = 0 \\
& = \lambda^2 \begin{vmatrix} \frac{\hbar^2}{\lambda} - \lambda & \frac{\sqrt{6}\hbar}{2} & 0 \\ \frac{\sqrt{6}\hbar}{2} & -\lambda & \frac{\sqrt{6}\hbar}{2} \\ 0 & \frac{\sqrt{6}\hbar}{2} & \frac{\hbar^2}{\lambda} - \lambda \end{vmatrix} = \lambda^2 [(\frac{\hbar^2}{\lambda} - \lambda)(-\hbar^2 + \lambda^2 - \frac{3}{2}\hbar^2) - \frac{\sqrt{6}\hbar}{2}(\frac{\sqrt{6}\hbar}{2}\frac{\hbar^2}{\lambda} - \frac{\sqrt{6}\hbar}{2}\lambda)] \\
& = \lambda^2 [(\frac{\hbar^2}{\lambda} - \lambda)(\lambda^2 - \frac{5}{2}\hbar^2) - \frac{3}{2}\frac{\hbar^4}{\lambda} + \frac{3}{2}\hbar^2\lambda] = \lambda^2 [(\lambda\hbar^2 - \frac{5}{2}\frac{\hbar^4}{\lambda} - \lambda^3 + \frac{5}{2}\hbar^2\lambda) - \frac{3}{2}\frac{\hbar^4}{\lambda} + \frac{3}{2}\hbar^2\lambda] \\
& = \lambda^2 (5\lambda\hbar^2 - 4\frac{\hbar^4}{\lambda} - \lambda^3)
\end{aligned}$$

$$\lambda_1 = 0; \lambda_2 = \hbar; \lambda_3 = -\hbar; \lambda_4 = 2\hbar; \lambda_5 = -2\hbar$$

$$\begin{aligned}
& \begin{bmatrix} -\hbar & \hbar & 0 & 0 & 0 \\ \hbar & -\hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -\hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & -\hbar & \hbar \\ 0 & 0 & 0 & \hbar & -\hbar \end{bmatrix} \times \vec{\xi}_1 = 0 \\
& \begin{bmatrix} \hbar & \hbar & 0 & 0 & 0 \\ \hbar & \hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & \hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & \hbar & \hbar \\ 0 & 0 & 0 & \hbar & \hbar \end{bmatrix} \times \vec{\xi}_{-1} = 0 \\
& \begin{bmatrix} 2\hbar & \hbar & 0 & 0 & 0 \\ \hbar & 2\hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & 2\hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & 2\hbar & \hbar \\ 0 & 0 & 0 & \hbar & 2\hbar \end{bmatrix} \times \vec{\xi}_{-2} = 0
\end{aligned}$$

$$\begin{bmatrix} -2\hbar & \hbar & 0 & 0 & 0 \\ \hbar & -2\hbar & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & -2\hbar & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & -2\hbar & \hbar \\ 0 & 0 & 0 & \hbar & -2\hbar \end{bmatrix} \times \vec{\xi}_2 = 0$$

$$\begin{bmatrix} 0 & \hbar & 0 & 0 & 0 \\ \hbar & 0 & \frac{\sqrt{6}}{2}\hbar & 0 & 0 \\ 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \frac{\sqrt{6}\hbar}{2} & 0 \\ 0 & 0 & \frac{\sqrt{6}\hbar}{2} & 0 & \hbar \\ 0 & 0 & 0 & \hbar & 0 \end{bmatrix} \times \vec{\xi}_0 = 0$$

$$\vec{\xi}_{-2} = \frac{1}{4} \begin{bmatrix} 1 \\ 2 \\ -\sqrt{6} \\ 2 \\ 1 \end{bmatrix}; \vec{\xi}_{-1} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ -1 \end{bmatrix}; \vec{\xi}_0 = \sqrt{\frac{3}{8}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{\xi}_1 = \frac{1}{2} \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}; \vec{\xi}_2 = \frac{1}{4} \begin{bmatrix} 1 \\ -2 \\ \sqrt{6} \\ -2 \\ 1 \end{bmatrix}$$

$$P_0 = \frac{1}{4}; P_1 = 0; P_2 = \frac{3}{8}; P_{-2} = \frac{3}{8}; P_{-1} = 0$$

SOLVED(WHY SO FUCKING LOOOOOOOOONG?????)